Harish Manoharan

Software engineer · agents, retrieval and evals

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EXPERIENCE

Purgo AI San Francisco Bay Area

AI startup: a LangGraph agent that turns data-engineering tickets into PySpark, SQL and dbt code, and validation products for regulated customers. I build and maintain the agent's prompts, workflow, retrieval and self-review layers.

Software Engineer Jun 2026 – Present

Software Engineer Intern Jun 2025 – May 2026

- Built the IQ/OQ compliance-validation engine life-sciences customers use to qualify Databricks on Azure and AWS for **signed audit reports**: 59 tests run as cancellable jobs with per-test timeouts, plus LLM-written tests from specs.
- Cut source tables missing from the agent's context **70%** (**79 → 24** on an eval set of 103 real customer tickets) by adding a dbt lineage-graph walk to its retrieval; chose the design by ablation, and no ticket regressed.
- Built the agent's Pinecone search over the Iceberg catalog: recall@10 **0.54 → 0.68** and latency **4.0 s → 0.18 s** vs Databricks Vector Search on 188 ticket-derived queries.
- Cut the cost of the agent's retrieval and drafting steps **7.6x**, holding recall within run-to-run noise across 5 runs, by moving them to a smaller model with per-step reasoning effort.
- Turned recurring Databricks runtime failures into AST lint rules that gate the agent's generated PySpark and SQL, plus a recovery pass that keeps the best candidate when every attempt fails.
- Shipped an LLM evaluation product that life-sciences customers use to audit models before use (IQ/OQ for LLMs): 10 behavioral tests on Azure OpenAI and Databricks-served models (hallucination, extraction F1, bias, determinism, latency) plus customer-defined tests; Next.js, NestJS, PostgreSQL.

Digitus Business Solutions Bengaluru, India

Software Engineer Intern May 2024 – Jul 2024

- Built a sales chatbot's text-to-SQL agent (schema lookup, retries): execution accuracy **52% → 76%** on 500 queries.
- Cut median helpdesk resolution time **70% (20 → 6 min)** with a cited RAG assistant over docs, videos and past tickets.

PwC India Bengaluru, India

Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Cut forecast error (MAPE) **15%** against the baseline across 50+ SKUs with XGBoost + SARIMA forecasts in Power BI.

PROJECTS

- **ReadableSTEM** (readablestem.com, open source, live): turns STEM PDFs, even scanned or handwritten, into accessible HTML; checks each equation against the PDF and escalates failures in tiers (cheap model, main model, instructor review) under spend caps. Pre-registered held-out evals on OpenStax sections and arXiv papers; on 15 arXiv papers, **8.4 silent errors per 100 equations** vs 13.7–16.3 for chatbots.
- **Margins** (studymargins.com, open source, live): turns a student's course files into cited notes, quizzes, practice exams and a day-by-day plan to the exam date (hybrid retrieval on pgvector).
- **Raising Kane** (3rd of ~40 teams, TestMu AI hackathon, Oct 2026): a coding agent that rebuilds a complex website's interface around what each visitor came to do, tested with Kane CLI flow simulations.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI: LangGraph, LangChain, MCP, RAG, vector search (Pinecone, pgvector), evals, LLM-as-judge, Langfuse, PyTorch

Backend, web, cloud: FastAPI, NestJS, React, Next.js, PostgreSQL, Prisma, AWS, Azure, GCP

Tools: AI coding agents (Claude Code, Codex), Git, GitHub Actions, Docker

EDUCATION

UT Dallas M.S. Computer Science · GPA 3.9/4.0 · Dean's Graduate Scholar May 2026

Coursework: Algorithms, Data Structures, Operating Systems, Database Design, Big Data Management

IIT Madras Dual Degree, M.Tech Data Science + B.S. Biological Sciences · INSPIRE Scholar (Govt. of India) May 2024